Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and the Babylonian Beru

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Code and data: `github.com/SethTenenbaum/gerizim-paper-a`

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Abstract

The UNESCO World Heritage Cultural and Mixed longitude distribution ($N = 1011$) contains a global 3° periodic structure: every longitude of the form x.18°E yields the maximum observed Tier-A+ count (sites within 6.7 km of a harmonic node, against a 4% geometric null). The 3° spacing equals 0.1 beru, the smallest integer sub-division of the Babylonian angular unit (MUL.APIN, c. 1000 BCE), and its phase falls within approximately 8 km of Mount Gerizim and within the same 35°E corridor as Jerusalem. The study begins with Mount Gerizim as anchor on the basis of its documented *tabbur ha-aretz* geographic-centrality tradition in Samaritan cosmography and the Levantine corridor's historical connection to Babylonian metrology. The global anchor sweep subsequently reveals that the x.18°E periodicity is a property of the UNESCO corpus itself, not of the specific choice of anchor within that corridor.

Domed and spherical monuments ($N = 90$, raw morphological keyword filter) show 12.2% Tier-A+ alignment against the 4% null (permutation $Z = 3.20$, $p = 0.004$; Bonferroni-adjusted $p_{\text{adj}} = 0.004$ across four primary tests). A leave-one-out check confirms that no single site drives this result: removing any one of the 11 A+ sites yields $N = 87$, A+= 10, 11.5%, 2.87×, $p = 0.0025$. Shifting the canonical spacing by only ±0.3% collapses significance in both the dome sub-population and the full corpus simultaneously. Harmonics containing at least one A+ site host 4.19× more total sites on average than harmonics with no A+ site (permutation $p = 0.0033$). The sites most strongly aligned with the grid are those UNESCO judged most foundational: the oldest capitals, the sacred origins, and the earliest inscription cohorts.

There is no obvious natural or geographic reason why cities and monuments across three continents and three millennia should cluster at 3° longitude intervals

measured from the Levant. The pattern is documented here so that it can be tested against independent corpora.

Keywords: archaeoastronomy, ancient metrology, geodetic alignment, UNESCO World Heritage, Babylonian beru, history of astronomy, statistical archaeology, spatial analysis, Mount Gerizim

1 Introduction

Prior studies of metrological patterns in monumental architecture have resisted rigorous evaluation because investigators assembled the site lists they then tested (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). This study starts from a different premise. It tests whether a site list the author did not compile and cannot modify shows a statistically unusual relationship to a specific angular interval attested in ancient Babylonian records.

The list is the UNESCO World Heritage Cultural and Mixed corpus ($N = 1011$). The angular interval is the 0.1-beru harmonic spacing (3°, derived from the *beru*, 30° of arc, MUL.APIN, c. 1000 BCE). The reference anchor is Mount Gerizim (35.269°E), selected on the basis of its documented role as the *tabbur ha-aretz* ("navel of the earth") in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989).

Testing the hypothesis reveals that the UNESCO longitude distribution contains a global 3° periodic structure whose phase is anchored to the Levantine corridor. Every longitude of the form x.18°E produces the maximum observed Tier-A+ count across the full 0–360° sweep, and the spacing between consecutive such anchors equals exactly 0.1 beru. This global periodicity is a property of the corpus itself; the choice of Gerizim as starting anchor falls within its phase band, but the same phase band is identically occupied by Jerusalem and any site in the ∼35°E corridor. The sites most strongly aligned with this grid are those UNESCO judged most foundational: ancient capitals, sacred origins, and sites inscribed in the earliest cohorts. There is no obvious natural or geographic reason why monuments across three continents and three millennia should cluster at 3° intervals measured from the Levant. Known sources of UNESCO selection bias produce geographic clustering but not regularly spaced longitudinal periodicity at sub-percent angular precision.

The analysis documents this pattern, quantifies it with a suite of statistical tests, and evaluates competing explanations (§2.4). It does not claim to resolve which explanation is correct. That resolution requires independent corpora beyond the scope of this paper.

The morphological sub-tests (§3.4, §3.5) ask whether domed, stupa, and mound structures show disproportionate spatial regularity above and beyond the full-corpus signal.

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), the composition of which is dated on broad grounds to the late second or early first millennium BCE. In this system, the unit divided the ecliptic into twelve equal segments and served for both stellar and terrestrial angular measurement. The terrestrial use is attested explicitly in MUL.APIN II ii 43–iii 15 (the "shadow-length" table), which converts beru of daylight into ground distances (Hunger & Steele, 2019, pp. 2–15). The sub-division of the beru into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to one-tenth of a *beru*, or exactly 3 *uš*, an integer number of the smallest standard angular unit.

The beru also functioned as a unit of territorial description in administrative and

cosmological contexts beyond astronomical tables. Royal inscriptions and administrative texts record inter-regional distances in beru between major cities and outlying territories (*nagû*) (Neugebauer, 1975; Robson, 2008). On the Babylonian *Imago Mundi* (the so-called Map of the World), the distances between the "islands" beyond the encircling Salt Sea are explicitly marked as multiples of beru, placing the unit in a cosmographic as well as a practical geographic register. The parallel institution of the *kudurru* (boundary stone) records land-grant limits and territorial extents in cuneiform inscriptions, and many kudurru were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987). The combination of a fractional angular unit employed at cosmological scales and a tradition of temple-deposited territorial records is consistent with the hypothesis that beru-based longitude spacing could have been embedded in the placement of major religious and civic foundations, though the present study tests the statistical pattern rather than the mechanism of transmission.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested scholarly history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique. Freeman showed Thom's quantum analysis was sensitive to selection effects (Freeman, 1976), and Kendall cautioned that apparent quanta can arise from rounding in the source data (Kendall, 1974). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference for the system within which the beru is embedded. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo permutation methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS–IAU thematic study (Ruggles & Cotte, 2010) provides the institutional context for archaeoastronomical analysis of World Heritage properties, and Aveni (Aveni, 2001, 2003) and Neugebauer (Neugebauer, 1975) supply the broader disciplinary framework.

The present study builds on this tradition but departs from it in three respects. First, the site list is externally curated (the UNESCO World Heritage List), eliminating the selection bias that has undermined many archaeoastronomical claims. Second, the test is one-dimensional (longitude only), yielding a well-defined geometric null expectation (4%) with no free parameters. Third, the analysis is fully computational and publicly archived, permitting exact reproduction (Tenenbaum, 2026).

Prior investigations of ley-line hypotheses (Watkins, 1925; Williamson & Bellamy, 1983; Hutton, 1991) were rightly criticized for post-hoc site selection and unconstrained geometric degrees of freedom (Kendall, 1974). The present design avoids all three problems: the site list is frozen before analysis, the test statistic is established in advance on metrological grounds, and the geometric null is analytically derived.

2.3 The Anchor: Mount Gerizim

The Levantine corridor near 35°E is the most naturally motivated starting point for a study connecting Babylonian angular measure to the broader ancient world. This is where the beru unit originated, where the oldest attested cosmographic traditions are located, and where the earliest long-distance trade and administrative networks of the ancient Near East converged.

Within this corridor, Mount Gerizim (35.269°E) was selected as the anchor because of its documented role as the cosmic center of the oldest continuous geographic-centrality tradition in the region. In Samaritan tradition Gerizim is the *tabbur ha-aretz* ("navel of the earth"), from which the earth was created and around which sacred geography is organized (Purvis, 1968; MacDonald, 1964; Crown, 1989). This tradition predates the major world religions whose sites appear in the UNESCO corpus and is analogous to the role Babylon and its meridian play in cuneiform astronomical records (Hunger & Steele, 2019). The selection was made on these textual grounds, and the analytic sequence is documented in the `git` commit history of the public repository (Tenenbaum, 2026).

The global anchor sweep (§5.3) tests all 36,000 longitudes from 0 to 360° and reveals a striking property of the UNESCO corpus itself: the top-scoring anchors are not randomly distributed, but cluster periodically at every $x.18$°E longitude, with 3° spacing between consecutive phase-matched anchors. Gerizim's longitude falls within this phase band, approximately 8 km from its nearest $x.18$°E node. This periodic structure is a property of the data, revealed by the sweep; it is not what motivated the original choice of anchor.

The central finding is therefore a property of the corpus, not of the specific anchor. Any site whose longitude falls near the $x.18$°E phase would score identically or near-identically as an anchor. Jerusalem (35.2317°E), 4.1 km from Gerizim, produces 56 A+ sites from the same corridor. Mecca reaches only $p \approx 0.057$ (ns). The result belongs to the 4.1 km corridor between Gerizim and Jerusalem, and the corridor comparison, while post-hoc, is the most anchor-invariant statement the data support.

The motivation for beginning in this corridor remains historically sound: the Levant is where the beru unit originated, where the oldest attested cosmographic traditions are located, and where the Samaritan geographic-centrality tradition places the cosmic reference point that this study uses as its anchor. That Gerizim occupies the same phase band as the global maximum is consistent with the metrological hypothesis, but it cannot be treated as independent evidence for it. The unit-sensitivity result (§5.1), in which ±0.3% shifts from the canonical spacing collapse significance in both the dome sub-population and the full corpus simultaneously, is the observation most resistant to any purely geographic account of the periodicity, because no known model of site density predicts sensitivity at sub-percent angular precision.

Gerizim is used throughout because its tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and not in the UNESCO XML test corpus, eliminating circularity by construction. All results are treated as exploratory: no timestamped pre-registration was filed, and the anchor was chosen prior to running the sweep, but the sweep's discovery of the $x.18$°E phase cannot be disentangled from the original anchor choice post hoc. The Bonferroni correction is retained as a conservative bound on false discovery; the dome enrichment ($p_{\text{adj}} = 0.004$) survives at $k = 4$ with substantial margin, and also at $k = 5$ (Table **??**).

2.4 Interpretive Framework

This paper is a pattern-detection study, not a historical attribution. The core observation is that the sites UNESCO ranked as most important are disproportionately likely to fall on a 0.1-beru harmonic grid anchored to the Levantine corridor. These are not a handful of sites in one region. They span every continent, every major religion, and three millennia of monument-building. The sites that UNESCO consistently judged most foundational tend to land near these longitude nodes at rates several times above chance, with the

strongest signal in the oldest inscription cohort.

Three classes of explanation are evaluated throughout. (1) **Geographic concentration.** The major civilizations that produced UNESCO-ranked monuments developed in Eurasian longitude bands that align with the 0.1-beru phase from the Levant. The simulation null models (§5.5) preserve the empirical longitude distribution; the dome sub-population survives all three, confirming the signal is not simply a consequence of site density at any given longitude. (2) **Metrological transmission.** A 3° (0.1-beru) angular reference system anchored near the Levant informed the placement of major foundations through trade, imperial administration, or religious networks. The unit-sensitivity collapse (±0.3% shifts destroy joint significance) and the morphological concentration in domed forms are consistent with this reading. (3) **Unmodeled statistical artifact.** A systematic bias in UNESCO curation, geocoding, or the discrete longitude distribution produces the apparent periodicity. An artifact account must explain why significance is unit-sensitive at ±0.3% precision, why the signal is concentrated in domed forms, and why it is strongest in the oldest inscription cohorts. None of these properties are predicted by any known curation or geocoding artifact.

The study documents the pattern and the evidence bearing on each explanation. Adjudicating between them requires independent evidence beyond this corpus.

3 Data and Methods

3.1 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, DMS E35 16 08, 2012). Gerizim is not on the inscribed World Heritage List and does not appear in the test-site corpus, avoiding circularity by construction. In each sweep the anchor is excluded from the test population; Jerusalem is also excluded in both sweeps (§2.3).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. Longitude differences are in angular measure; the tier classification is therefore a fixed angular criterion, not a fixed physical radius. The kilometre labels reported alongside tier thresholds (e.g. $\lesssim 6.7$ km for Tier-A+) are approximate equatorial equivalents computed as $\delta \times 30 \times 111$ km/° and become smaller at higher latitudes: a site at 60°N that satisfies $\delta \leq 0.002$ beru is physically within $\sim 55.7/111 \times 6.7 \approx 3.4$ km east-west of the harmonic. The Gerizim–Jerusalem separation of 4.1 km is likewise an equatorial approximation (no cosine-latitude correction); at 32°N the true east-west distance is approximately $4.1 \times \cos(32) \approx 3.5$ km. This convention is applied uniformly throughout the study so that all reported kilometre values are directly comparable. This makes the angular criterion *conservative* relative to a fixed physical-km threshold at high latitudes. A geodesic sensitivity check (east-west haversine distances) confirms this: 0 of the 56 current A+ sites would be removed under a strict 6.7 km physical threshold, while 14 additional high-latitude sites would qualify, increasing the A+ count to 70 and strengthening all primary results. No correction for latitude is therefore applied; the reported km figures are equatorial approximations provided for descriptive context only. The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$. A Gaussian KDE fitted to the observed longitude distribution produces 40.4 ± 6.2 expected A+ out of 1011 sites (∼4.0%), confirming the geometric null is not inflated by longitude clumping (§5.5).

Throughout this paper, exact binomial p-values are reported as a transparent reference

baseline against the geometric null. The *primary* tests against geographic structure are the simulation-based null models (§5.5), including longitude permutation, bootstrap, and KDE-smoothed null, which preserve the empirical longitude distribution and do not assume i.i.d. sampling. Where simulation and binomial p-values diverge, the simulation figure takes precedence. Binomial p-values are best understood as convenient approximations that assume uniform longitude spacing. Because this assumption is violated by the UNESCO corpus, the simulation models are designed to relax it.

3.2 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested throughout this paper is 0.1 beru (3° of arc), not a free parameter but the smallest integer sub-division of the beru attested in cuneiform metrology (§2.1). The 0.1-beru interval equals exactly 3 *uš* (3°), an integer multiple of the smallest standard angular unit in the sexagesimal system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). No alternative spacing was considered or tested before the primary analysis, and the choice is established on metrological grounds alone.

The empirical unit sensitivity sweep (§5.1, Tables 11–12) serves as verification that the metrological basis is empirically supported, as it is the only spacing that produces joint significance in both the domed/spherical monument population and the full UNESCO corpus. A -0.5% shift from the canonical value collapses dome significance from $p = 0.0005$ to $p > 0.11$, and a $\pm 0.3\%$ shift collapses joint significance in both populations simultaneously. Adjacent spacings (0.06, 0.07, 0.09, 0.12 beru) are non-significant in both populations. This sweep corroborates the metrological argument empirically and is not claimed as an independent route to the same conclusion.

Non-independence of the anchor sweep and unit sensitivity test. The global anchor sweep (§5.3) and the unit sensitivity sweep (§5.1) both probe the same underlying $x.18$°E periodic structure in the UNESCO longitude distribution. The anchor sweep reveals the 3° periodicity by scanning all trial longitudes, and the unit sensitivity sweep confirms that the 0.1-beru spacing produces the peak signal. These are two views of the same pattern and should not be treated as independent lines of evidence. Independent evidence comes from the morphological specificity (dome enrichment surviving simulation nulls) and the temporal gradient (which does not depend on the anchor sweep or the unit sweep).

3.3 Tier Classification (fixed before data analysis)

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic (interval $\approx 3°$). All tier boundaries and the six test populations were defined and frozen before any beru values were computed. The corpus was not compiled by the author and cannot be modified.

- **Tier A++**: $\delta \leq 0.0002$ beru ($\lesssim 0.67$ km at equator). $P_0 = 0.4\%$. Exploratory; reported descriptively.
- **Tier A+**: $\delta \leq 0.002$ beru ($\lesssim 6.7$ km at equator). $P_0 = 4\%$. Primary test threshold.[1]
- **Tier A**: $\delta \leq 0.010$ beru ($\lesssim 33$ km at equator). $P_0 = 20\%$.

[1] $2 \times 0.002/0.1 = 4\%$: window width divided by harmonic spacing. Km labels are equatorial approximations; see §3 for the latitude-sensitivity discussion.

- **Tier B**: $\delta \leq 0.050$ beru.

3.4 Domed and Spherical Monuments (Test 2)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV web-page text. All 7 keywords are treated identically (positive whole-word match, no context gating), making the population mechanically determined by the keyword list. Final population: $N = 90$. Primary tests: simulation-based null models (permutation, bootstrap, KDE) reported in §5.5; exact binomial A+ ($> 4\%$) and A ($> 20\%$) reported as transparent reference baselines; χ^2 uniformity and anchor sweep also reported.

A context-validated robustness check (Test 2x) applies sentence-level architectural confirmation. Gate 1 (28 negative patterns) rejects geological, vernacular, and non-architectural uses, while Gate 2 (86 positive patterns) requires architectural confirmation. This reduces the population to $N = 83$ (7 sites rejected, e.g. Uluru "rock domes," Hiroshima "Genbaku Dome," full list in Appendix A.1). The validated version yields $p = 0.0005$, $3.31\times$ enrichment, which is stronger than the raw sweep ($p = 0.0009$, $3.06\times$), confirming that the 7 rejected sites are non-architectural matches diluting the signal, not driving it. Because the validation rules were developed after the initial keyword sweep, Test 2x is a post-hoc robustness check and is excluded from the primary Bonferroni family. The raw-sweep result is the conservative primary figure.

3.5 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends the core domed/spherical test (Test 2) by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun, mound*). The categories are grouped on the basis of shared hemispherical or quasi-hemispherical geometry, a form widely associated with cosmological and funerary symbolism across cultures. They are not grouped on the basis of a claimed architectural lineage. The label "evolution" refers to a morphological sequence in the statistical test design (mound $\rightarrow$ stupa $\rightarrow$ dome as progressively formalized built expressions of the same form class); it is not a claim that any category is architecturally derived from another.

The primary version (Test 2b) applies a raw positive keyword sweep: all 132 Cultural/Mixed sites matching any keyword in {*tumulus, tumuli, barrow, barrows, kofun, mound, stupa, stupas, dagoba, chorten, tholos, dome, domed, domes, spherical*} are included without context gating, for the same pre-specification reasons as Test 2. A context-validated version (Test 2bx, Exploratory) applies the same architectural gates as Test 2x, reducing the population to $N = 126$. The validated version yields $p = 0.0005$, $2.78\times$ enrichment, which is stronger than the raw sweep ($p = 0.0009$, $2.65\times$), again demonstrating that the extra sites in the raw sweep are diluting rather than driving the signal.

Test 2b is classified as a *sensitivity variant* of Test 2 rather than an independent hypothesis. Its population shares the entire dome/stupa core ($N = 90$) with Test 2 and adds only 42 mound-stage sites, which show no significant enrichment on their own (6.2%, $p = 0.3012$, ns). Counting Test 2b separately in the Bonferroni family would add a test whose population is almost entirely overlapping with Test 2. Readers who prefer a stricter $k = 5$ family should note that the dome result ($p_{\text{adj}} = 0.004$ at $k = 4$, $p_{\text{adj}} = 0.005$

at $k = 5$) survives either way. Tests: exact binomial A+ on combined and per-stage populations.

3.6 Cluster Asymmetry (Test 3)

Full corpus ($N = 1011$), no keyword filter. If the beru-harmonic hypothesis is correct, ancient surveyors would have preferentially founded or selected significant sites at harmonic longitudes, and subsequent construction, urban growth, and heritage accumulation would have clustered around those initial seed sites. This predicts that Tier-A+ harmonics should host more UNESCO sites in total, not just more A+ sites, than harmonics without any A+ hit. For each site, count other UNESCO sites within $\pm 0.3°$ longitude. Compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

3.7 Founding-Site Enrichment (Test E [Exploratory])

Two independent keyword analyses on the full corpus ($N = 1011$), no keyword filtering for the primary tests:

1. **World religion founding sites.** Keyword match for Buddhism, Christianity, Islam, Judaism, Hinduism, Zoroastrianism in site name/XML description.

2. **UNESCO founding language.** Sites where UNESCO uses "first capital," "birthplace of," "founded in," etc. in the XML description.

A context-aware keyword classifier (`lib/founding_filter.py`) was applied to all sites. This classifier was iteratively tuned after examining misclassified sites (full disclosure in §3.10). Because refinement occurred after data inspection, the enrichment result from this classifier should be treated as *Exploratory.* The inscription-year gradient and world-religion sub-tests do not depend on this classifier.

3.8 Temporal Gradient (Test 4)

Cochran-Armitage trend test on the full corpus ($N = 1011$), no keyword filter. A+ rate is tested for a monotonic decline as UNESCO inscription year increases (earliest inscriptions reflect greatest site importance). Three-cohort grouping (1978–1984, 1985–1999, 2000–2025) is primary; five-cohort grouping (1978–1984, 1985–1991, 1992–1999, 2000–2009, 2010–2025) is secondary. The three-cohort bins are formed by merging the two middle five-cohort bins (1985–1991 and 1992–1999) into a single 1985–1999 bin. Code: `analysis/unesco/temporal_gradient_test.py`.

Note on the regional composition confound. The founding canon (1978–1984) is concentrated in Europe and the Mediterranean corridor, the Eurasian heartland where the beru-harmonic signal is strongest. Later cohorts add proportionally more sites from Africa, the Americas, and Oceania, regions where the null hypothesis predicts lower A+ rates regardless of any metrological effect. A within-region diagnostic (`regional_temporal_gradient.py`) finds that 4 of 5 regions show the early greater than late direction, but 0 reach individual significance, and the diagnostic is underpowered at $N = 5$ regions. The Mantel-Haenszel region-adjusted OR is 1.582 ($p = 0.1101$). The temporal gradient is therefore treated as descriptive only; it indicates that UNESCO's

earliest inscription cohort is concentrated in the very regions where the beru-harmonic signal is strongest, but does not establish that time itself, independent of geography, drives the pattern.

3.9 Multiple-Comparisons Protocol

Four test populations were identified before any beru distances were computed (Tests 1, 2, 3, 4); however, because no timestamped pre-registration was filed, all results are treated as exploratory throughout. The Bonferroni correction is retained as a conservative familywise bound and is reported alongside simulation-based p-values, which are the primary test against the chance explanation. The Bonferroni family size is $k = 4$, corresponding to the per-test threshold $\alpha/k = 0.0125$.

Test E (founding-site classifier enrichment) is excluded from the Bonferroni family because the classifier (`lib/founding_filter.py`) was iteratively refined after data inspection, so its enrichment result is post-hoc (§3.10). Test 2b is classified as a sensitivity variant of Test 2 and is not counted separately. The population of Test 2b shares the entire dome and stupa core ($N = 90$) with Test 2 and adds only 42 mound-stage sites, which show no significant enrichment on their own (6.2%, $p = 0.3012$, ns). Adding a test whose population almost entirely overlaps an existing family member would inflate the family count and dilute the correction without adding an independent test of a distinct hypothesis. Readers who prefer a stricter $k = 5$ family should note that the dome result ($p_{\text{adj}} = 0.004$ at $k = 4$, $p_{\text{adj}} = 0.005$ at $k = 5$) survives either way.

Table 1: Bonferroni correction under two family-size assumptions. $k = 4$ counts Tests 1, 2, 3, 4 as the primary family. $k = 5$ adds Test 2b as an independent test. The dome result survives correction under both assumptions. All results are exploratory; the correction is a conservative familywise bound, not a pre-registration claim.

Test	Raw p	p_{adj} ($k = 4$)	p_{adj} ($k = 5$)	Survives ($k = 5$)
Test 1 (full corpus A+)	0.0103	0.041 *	0.052 †	No (marginal)
Test 2 (dome/spherical A+)	0.0009	0.004 **	0.005 **	Yes
Test 2b (hemispherical evol.)	0.0009	—	0.005 **	Yes
Test 3 (cluster asymmetry)	0.0033	0.013 *	0.017 *	Yes
Test 4 (temporal gradient)	0.0481	0.192 ns	0.241 ns	No

Per-test threshold: $k = 4$: $\alpha = 0.0125$; $k = 5$: $\alpha = 0.01$.

Test dependence. Tests 1 and 3 share the 56 A+ sites that drive both signals, making them partially dependent. A strict family-wise correction for these two tests would treat them as drawing from the same underlying signal rather than as two independent tests. The $k = 4$ Bonferroni correction does not account for this dependency and is therefore somewhat liberal for the Test 1 / Test 3 pair. The result that carries primary weight is Test 2 (dome/spherical, $p_{\text{adj}} = 0.004$), which uses a morphologically filtered sub-corpus independent of Tests 1 and 3.

Test 1 (full corpus, $p_{\text{adj}} = 0.041$) survives correction at $k = 4$, though the margin is modest. A methodological asymmetry bears on the interpretation. Gerizim is an external anchor not present in the 1011-site inscribed corpus, so its own longitude does not contribute a data point when counting quantized pairs. Jerusalem is an inscribed corpus member and is self-excluded from its own evaluation ($N = 1010$). Both anchors

produce the same marginal result ($p \approx 0.010$) and the corridor between them (the 4.1 km window that produces 56 A+ sites from either vantage point, §5.3) is the more robust finding, although the corridor comparison itself is post-hoc. Test 4 (temporal gradient, $p_{\text{adj}} = 0.192$) does not survive correction. A within-region diagnostic further shows the pooled gradient is largely explained by regional composition (4/5 regions concordant, MH OR = 1.582, $p = 0.1101$; §4.6). Test 4 is treated throughout as *descriptive only*.

Results surviving Bonferroni correction are dome/spherical ($p_{\text{adj}} = 0.004$), full corpus enrichment ($p_{\text{adj}} = 0.041$), and cluster asymmetry ($p_{\text{adj}} = 0.013$). A comprehensive FDR audit (43 tests, Benjamini & Hochberg 1995) retains 24 at $q < 0.05$, of which 6 survive Bonferroni at $k = 43$. All data, code, and random seeds are publicly archived (Tenenbaum, 2026).

3.10 Keyword Filter Methodology and Post-Hoc Tuning Disclosure

The dome/stupa filter (§3.4) required minimal iteration: the keyword set follows directly from architectural typology and is effectively fixed in advance of any beru computations. The founding-site filter (`lib/founding_filter.py`) was developed iteratively by examining misclassified sites. All corrections used domain-level vocabulary, with no site-specific exclusions and no longitude-based decisions at any stage. The founding-enrichment result ($p = 0.009$) is directly affected by this tuning. The three Bonferroni-surviving results (Tests 1, 2, 3) and the keyword-independent sub-tests of Test E and Test 4 (full-corpus enrichment, dome population, hemispherical evolution, inscription-year gradient, world-religion keywords, corridor precision, simulation nulls) are not affected by the founding classifier in any way. Full keyword lists, gate logic, and per-site audit trails are deposited as supplementary material (Appendix A).

3.11 Null Model Validation

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each): (1) longitude permutation, (2) bootstrap from empirical distribution, (3) KDE-smoothed null. Results in §5.5.

As an independent external control, the UNESCO corpus is compared against the Wikidata P1435 global heritage corpus (property `P1435` = "heritage designation"; $N = 62{,}785$ globally-distributed structures fetched via the Wikidata Query Service in April 2026). Unlike the UNESCO corpus, the P1435 dataset includes structures of all significance levels and is geographically broader. It serves as a baseline to test whether UNESCO enrichment reflects generic heritage-site distribution rather than a corpus-specific signal. Code: `data/scripts/fetch_p1435_global.py`.

4 Results

Results are presented in test order (Tests 1, 2, 2b, 3, 4, E). The simulation-based null models (§5.5) serve as the primary test of the chance explanation.

4.1 Global Enrichment (Test 1)

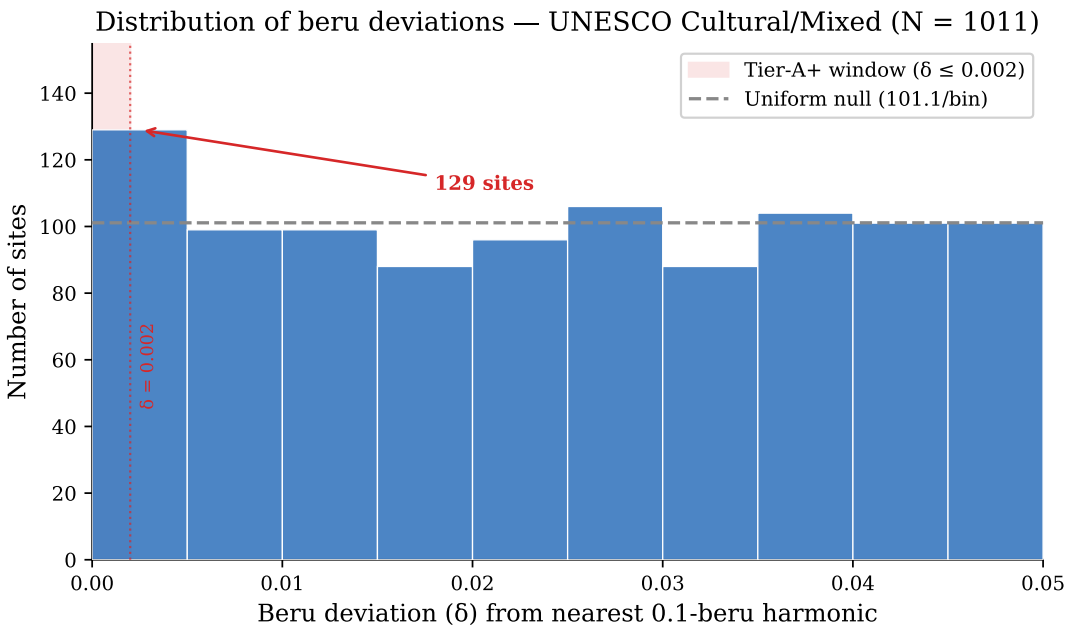

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) is the primary result.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), significantly above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.041$, marginal). Gerizim (external, not in the corpus) and Jerusalem (inscribed, self-excluded to $N = 1010$) both produce this marginal result ($p = 0.010$). The signal is a property of the ∼35°E corridor, not of either anchor individually (§5.3). This keyword-independent result is the primary finding. The full-corpus enrichment ratio (1.38×) is substantively modest. A 5.5% rate against a 4% null means approximately 15 additional A+ sites above chance expectation in a corpus of 1011. This is consistent with a scenario in which only a fraction of monumental sites were placed with reference to a grid, and the signal is diluted by the majority that were not. The dome sub-population (3.06×) and stupa sub-population (5.56×) show stronger effect sizes. The sub-tests below characterize the signal by site type, cluster structure, founding importance, and temporal ordering.

Region-stratified robustness. A Mantel-Haenszel analysis stratified by UNESCO region yields OR = 1.582, $p = 0.1101$ (trend-level). When tested against region-specific empirical null rates, enrichment remains significant ($p = 0.0013$; pre-2000: $p = 0.0076$). The enrichment is not an artifact of geographic concentration.

4.2 Domed and Spherical Monuments

Applying the 7-keyword morphological filter to the 1248-entry UNESCO XML yields $N = 90$ sites (raw positive sweep; no context gating).

The primary tests against the chance explanation are the simulation-based null models (§5.5): permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$. Exact binomial results

are also reported as transparent reference baselines against the geometric null. The non-significant χ^2 ($p = 0.0732$) is not in tension with the binomial ($p = 0.0009$): the hypothesis predicts a narrow spike at harmonics, not a uniform gradient, and a five-bin χ^2 has poor power to detect such a concentrated departure at $N = 90$. The context-validated robustness check (Test 2x; $N = 83$, 7 sites rejected) yields an equal-or-stronger signal ($p = 0.0005$, 3.31× enrichment), confirming that the 7 rejected sites are diluting rather than driving the result.

Table 2: UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). Primary evidence against the chance explanation: simulation permutation $Z = 3.20$, $p = 0.004$ (§5.5). Exact binomial results are reference baselines assuming uniform longitude spacing. The non-significant χ^2 is discussed in the text.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ ($\leq$ 0.002 beru)	11	3.60	12.2% [6.3, 20.8]	3.06×	0.0009	***
Tier-A ($\leq$ 0.010 beru)	29	18.00	32.2% [22.9, 42.7]	1.61×	0.0043	**
χ^2-uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0: A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 Mean δ: 0.0220 beru
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, permutation $p = 0.004$, Bonferroni $p_{\text{adj}} = 0.004$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini, the Birthplace of the Lord Buddha (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an–Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km). Individual-site p-values do not survive correction, and evidential weight derives from the population-level tests. Note that Lumbini is in the Test 2 corpus through the *stupas* keyword match; its classification as a birthplace site rather than a domed monument is addressed in §A.1. A leave-one-out check confirms the result is not driven by any single site: removing any one of the 11 A+ sites yields $N = 87$, A+= 10, 11.5%, 2.87×, $p = 0.0025$ (same for all 11 removals).

In the global anchor sweep (§5.3), Gerizim scores 55 A+ (98.3th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only 2.4% of 1741 candidate words achieve comparable enrichment.

4.3 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

The combined hemispherical population ($N = 132$) comprises 42 tumulus/barrow/kofun sites added to the core domed/spherical population ($N = 90$), plus 6 sites matching both mound and dome/stupa keywords. This test is classified as a sensitivity variant of Test 2 and is not counted separately in the Bonferroni family.

Table 3: Test 2b: hemispherical mound evolution ($N = 132$). Combined tumulus → stupa → dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ (≤ 0.002 beru, ≤ 6.7 km)	14	4.16	10.6%	0.0009	***
Tier-A (≤ 0.010 beru, ≤ 33 km)	41	20.80	31.1%	0.0017	**

Expected under H_0: A+ = 4%, A = 20%. Enrichment: A+ = 2.65×, A = 1.55×.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

Stage-by-stage breakdown (Table 3):

Table 4: Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	48	3	6.2%	1.55×	0.3012 ns
Stupa (stupa/stupas)	18	4	22.2%	5.56×	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	2.68×	0.0102 *
Combined	**132**	**14**	**10.6%**	**2.65×**	**0.0009 *** **

The combined population ($N = 132$, A+ = 10.6%, $p = 0.0009$ ***) is consistent with the Test 2 result. The mound stage itself does not reach significance (6.2%, $p = 0.3012$, ns), and its inclusion reduces the combined A+ rate from the 12.2% of Test 2 alone to 10.6%. The signal in Test 2b is carried by the stupa and dome sub-stages, confirming that the Test 2 finding is robust to boundary expansion of the keyword set. The three mound-stage A+ sites (Troy, $\delta = 3.3$ km, ∼3000 BCE; Takht-e Soleyman, $\delta = 3.8$ km; Gaya Tumuli, $\delta = 5.8$ km, 1st–6th c. CE) are individually notable, and the stage-level rate (1.55×) is in the predicted direction, but the corpus is too small to produce a significant result. The stupa stage (5.56×, $p = 0.0050$ **) shows the highest enrichment of any stage. The small corpus ($N = 18$) warrants caution, but the result is consistent with the dome/spherical signal in Test 2 rather than dependent on it, since the stupa keyword set is disjoint. A leave-one-out check confirms no single stupa site drives the result: removing any one of the 4 A+ stupa sites yields $N = 17$, A+= 3, 17.6%, $p = 0.0286$ (same for all 4 removals).

4.4 Cluster Asymmetry

Table 5: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.3°$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.3°$ lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier ≤ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		4.19×	< 0.0001 ***

A+-bearing harmonics host 4.19× more sites on average than non-A+ harmonics ($p < 0.0001$). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.0238$).

4.5 Founding-Site Enrichment

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011), above the 4% null (binomial $p = 0.0103$ *, enrichment 1.38×, Bonferroni-adjusted $p_{\text{adj}} = 0.041$, marginal).

Line 1: World religion founding sites

Sites associated with any world religion ($N = 363$) show 6.1% Tier-A+ (22/363, 1.52×, $p = 0.0369$ *). Christianity ($N = 219$) shows 6.8% A+, $p = 0.0312$ *. Buddhism ($N = 46$ sites with founding-religion keyword match, noting that the broader Buddhist heritage corpus of $N = 95$ is analyzed separately): 8.7% A+ ($p = 0.11$, ns). Jerusalem falls within the Tier-A+ window at 4.1 km.

Line 2: Inscription-year stratification and formal trend test

Table 6: Tier-A+ enrichment by UNESCO inscription-year cohort ($N = 1011$ Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)	138	11	8.0%	1.99×	0.0232 *
pre-2000 (1978–1999)	501	33	6.6%	1.65×	0.0041 **
2000–2025 (modern era)	510	23	4.5%	1.13×	0.3081 ns
ALL (1978–2025)	1011	56	5.5%	1.38×	0.0103 **
Fisher exact (1978–1984 vs. 2000–2025): OR = 1.83, $p = 0.0844$ (†)					

The A+ rate is highest in the founding canon (1978–1984: 8.0%, 1.99×, $p = 0.0232$) and dilutes to the null in the modern era.

Five-cohort gradient (Table 6). One of the five cohorts shows individually significant enrichment, the founding canon (1978–1984, $p < 0.05$). The 1992–1999 cohort is marginal ($p = 0.064$), and the remaining three cohorts do not reach significance. The 1985–1991 cohort breaks the otherwise monotonic decline (see note below Table 6 for explanation). The two post-2000 cohorts are indistinguishable from the 4% null.

Table 7: Five-cohort Tier-A+ gradient: UNESCO inscription year ($N = 1011$). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984	138	11	8.0%	1.99×	0.023 *
1985–1991	141	8	5.7%	1.42×	0.205 ns
1992–1999	222	14	6.3%	1.58×	0.064 ∼
2000–2009	211	11	5.2%	1.30×	0.227 ns
2010–2025	299	12	4.0%	1.00×	0.536 ns

The broad pattern is clear. The earliest cohort shows individually significant enrichment ($p < 0.05$), and the 1992–1999 cohort is marginal. Post-2000 cohorts are indistinguishable from the 4% null.

Note on the 1985–1991 dip. This cohort (1.42×, ns) interrupts the monotonic decline, consistent with a disproportionate share of Mixed-category sites in this period (7.8% of cohort entries vs. 2.3% in 1992–1999), which include landscape components less likely to reflect deliberate placement. The three-cohort grouping, which merges this period with 1992–1999, is established prior to the analysis and robust to within-period variation.

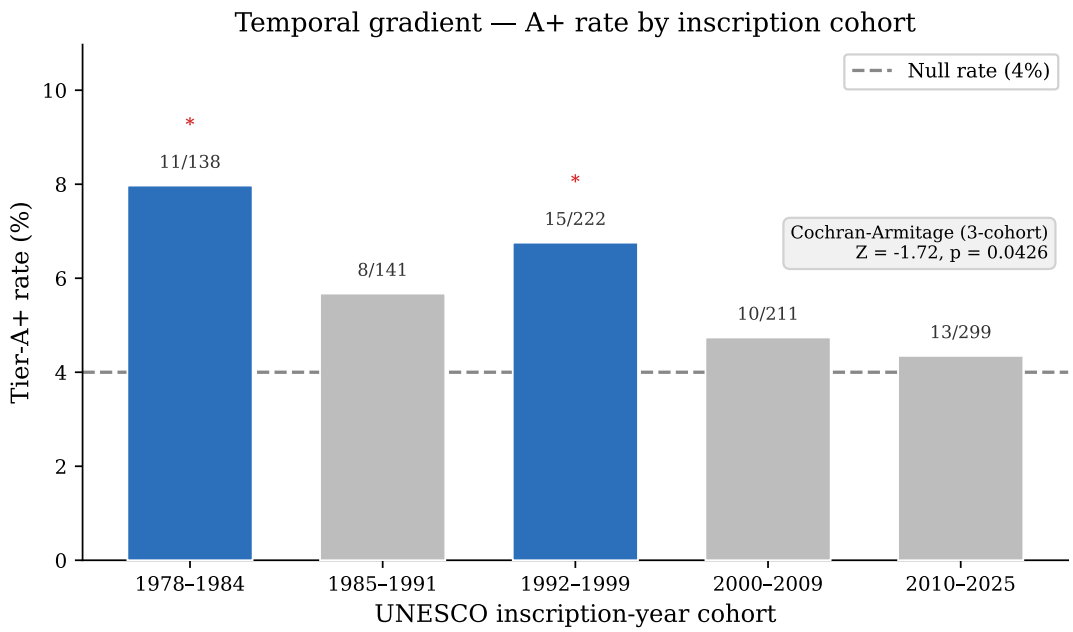

Figure 2: Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows ∼8% A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test: $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided).

Cochran-Armitage trend test.

- **Three-cohort:** $Z = -1.66$, $p = 0.0481$ (*), one-sided.
- **Five-cohort:** $Z = -1.66$, $p = 0.0489$ (*), one-sided.

Regional composition confound and within-region diagnostic. As noted in §3.8, the founding canon (1978–1984) is concentrated in Europe and the Mediterranean while later cohorts draw more heavily from Africa, the Americas, and Oceania. A region-stratified diagnostic (`regional_temporal_gradient.py`) tests whether the gradient persists within individual regions. Of 5 UNESCO regions, 4 show the early > late direction, but 0 reach individual significance ($p < 0.05$, one-sided Fisher). The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). The temporal gradient is largely explained by regional composition after adjustment and is treated as *descriptive only* throughout.

Running-cumulative stability. The cumulative A+ rate reaches $p < 0.05$ by $N = 103$ (through 1982) and remains significant ($p < 0.05$) through the full corpus. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003), identified post-hoc (Table 7).

Table 8: Sequential significance: cumulative A+ enrichment as the UNESCO corpus grows year by year. The corpus is reconstructed in the order UNESCO inscribed it. Peak $-\log_{10}(p)$ occurs at $N = 605$ (through 2003); full-corpus significance persists but at reduced strength. Code: `deep_temporal_analysis.py`.

Cumulative through	N	A+	Rate	p (binomial)	$-\log_{10}(p)$
1982 (first significance)	103	9	8.7%	0.023 *	1.65
1984 (founding canon)	138	11	8.0%	0.0232 *	1.63
1993	324	24	7.4%	0.003 **	2.50
1999 (pre-2000 complete)	501	33	6.6%	0.0041 **	2.39
2003 (PEAK)	**605**	**43**	**7.1%**	**0.00025926 (***)**	**3.59**
2010	728	44	6.0%	0.005 **	2.29
2017	864	49	5.7%	0.010 *	1.98
2025 (full corpus)	1011	56	5.5%	0.0103 *	1.99

Enrichment is concentrated in the first half of the corpus (≤2000: 6.5%, $p = 0.004581$; 2001–2025: 4.5%, $p = 0.295198$ ns). The sequential peak is post-hoc and classified as exploratory.

Line 2b: Founding-era dating from UNESCO descriptions

Founding-era estimates were extracted from UNESCO XML `short_description` texts for 579 of 1011 sites (code: `deep_temporal_analysis.py`). The BCE combined bin (173 sites, 9 A+, 5.2%) shows directional elevation above the null rate across both sub-epochs, consistent with the founding-class hypothesis prior to any modern surveying context ($p = 0.2577$, ns; underpowered given N). The early-modern bin (1500–1800 CE: 7.9%, $p = 0.0303$ *) is the only individually significant era, driven by founding colonial capitals (e.g., Antigua Guatemala, Grand-Bassam, Mbanza Kongo); this likely amplifies an existing pattern rather than generating it, given the directional consistency seen across earlier bins. The classical-antiquity bin (1000 BCE–1 CE) is elevated relative to the null rate (1.85×, $p = 0.106$, ns, $N = 81$).

The nine BCE A+ sites span three continents and at least five independent cultural traditions, ranging from Bronze Age urban foundations (Troy, ∼3000 BCE; Mycenae/Tiryns,

2nd mill. BCE) to classical-era state monuments (Via Appia, 312 BCE; Great Wall, 220 BCE). This suggests the pattern is not an artifact of any single tradition or region. Sites inscribed in the founding canon (≤1984) with post-CE founding dates show 15.1% A+ ($p = 0.0012$ **, $N = 53$).

Table 9: Tier-A+ rate by estimated founding era (extracted from UNESCO descriptions). Only the 579 sites with extractable dates are included. Null rate $P_0 = 4\%$.

Era	N	A+	Rate	Enrichment	p (binomial)
Deep antiquity (pre-1000 BCE)	92	3	3.3%	0.82×	0.717 ns
Classical antiquity (1000 BCE–1 CE)	81	6	7.4%	1.85×	0.106 ns
BCE combined	**173**	**9**	**5.2%**	**1.30×**	**0.2577 ns**
Late antiquity (1–500 CE)	30	2	6.7%	1.67×	0.339 ns
Medieval (500–1500 CE)	175	10	5.7%	1.43×	0.165 ns
Early modern (1500–1800 CE)	126	10	7.9%	1.98×	0.0303 *
Modern (post-1800 CE)	75	2	2.7%	0.67×	0.807 ns

The A+ rate is elevated relative to the null across most eras (Table 8), indicating the pattern is not exclusively a product of early-modern selection. The deep-antiquity bin (<1000 BCE) is near the null rate, as expected: pre-1000 BCE sites are overwhelmingly continuous landscapes rather than discretely founded monuments. The key distributional finding is that *inscription year* is a stronger predictor of A+ status than founding era. Canon-inscribed (≤1984) post-CE sites show 15.1% A+ ($p = 0.0012$ **, $N = 53$), the highest rate of any subgroup. The nine BCE A+ sites (listed above) are all founding-class monuments.

Line 3: UNESCO's own founding language

Sites where UNESCO texts use founding/primacy language show elevated A+ rates. "Founded in" (57 sites, 5 A+, 8.8%, $p = 0.077$) and "first … church" (4 sites, 2 A+; $n = 4$ is too small for a meaningful p-value and is noted descriptively only) both show elevated rates.

Thematic catalog of all 56 Tier-A+ sites

Keyword classification (Table 9) identifies 25/56 (45%) A+ sites as founding-class, and 28 (50.0%) match no keyword. The keyword enrichment test yields OR = 2.04, $p = 0.009$ *, and keyword lists were iteratively refined (§3.10). Top 3 most precise hits: Sacri Monti ($\delta = 0.000024$ beru, 0.1 km), Khoja Ahmed Yasawi ($\delta = 0.000069$ beru, 0.2 km), Site of Palmyra ($\delta = 0.000078$ beru, 0.3 km). All are selected post-hoc and reported descriptively.

Table 10: Thematic classification of all 56 Tier-A+ sites in the full UNESCO Cultural/Mixed corpus ($N = 1011$). Categories assigned from UNESCO inscription text. Full catalog in `founding_sites_analysis.py`.

Category	N	% of A+
Founding Capital / Seat of Power	11	19.6%
Founding Monument / Prototype	6	10.7%
Sacred Origin / Birth of a Tradition	4	7.1%
Founding Axis / Imperial Infrastructure	4	7.1%
Combined "Founding" (F+S+M+X)	**25**	**45%**
Ancient Continuous Landscape	3	5.4%
Unclassified (no keyword match)	28	50.0%

Convergence summary

Table 11: Test E convergence. Independent predictors of "founding importance" all identify the same A+ enrichment pattern. Rows are grouped by analysis status (horizontal rules as visual separators), where **Pre** = population defined before any beru distances were computed (strongest methodological standing, but not formally pre-registered), **Post** = post-hoc (decided after seeing data but before reporting), and **Expl** = exploratory (identified in data, reported descriptively only). Because no timestamped pre-registration was filed, all results are treated as exploratory. Post/Expl rows are supporting evidence only.

Predictor	N	A+ rate	Enrichment	p	Status
Population defined before beru distances were computed (strongest methodological standing)					
Christian sacred sites	219	6.8%	1.71×	0.0312 *	Pre
Any world religion (union)	363	6.1%	1.52×	0.0369 *	Pre
Earliest inscribed (1978–1984)	138	8.0%	1.99×	0.0232 *	Pre
Pre-2000 inscriptions	501	6.6%	1.65×	0.0041 **	Pre
Full corpus (no filter)	1011	5.5%	1.38×	0.0103 **	Pre
Modern additions (2000+)	510	4.5%	1.13×	0.3081 ns	Pre
Post-hoc (supporting evidence only)					
First half (≤2000)	505	6.5%	1.68×	0.004581 **	Post
Second half (2001+)	506	4.5%	1.14×	0.295198 ns	Post
Exploratory (descriptive only; no confirmatory weight)					
Peak sequential (≤2003)	605	7.1%	1.78×	0.00025926 ***	Expl
Cochran-Armitage trend (3 cohorts): $Z = -1.66$, $p = 0.0481$ (*) [Pre]					
Sequential peak at $N = 605$: $p = 0.00025926$ (***); full-corpus $p = 0.0103$ is 10× weaker [Expl]					

The signal is present in the full corpus ($p = 0.0103$), concentrated in religion-associated sites ($p = 0.0369$) and the earliest-inscribed cohort ($p = 0.0232$), and absent in the

modern era ($p = 0.3081$ ns). The inscription-year trend (Cochran-Armitage $Z = -1.66$, $p = 0.0481$) is reported as the formal Test 4 (§4.6).

4.6 Temporal Gradient (Test 4)

The Cochran-Armitage trend test (no keyword filter) yields $Z = -1.66$, $p = 0.0481$ (three-cohort, one-sided, five-cohort $Z = -1.66$, $p = 0.0489$). Neither survives Bonferroni correction ($p_{\text{adj}} = 0.192$, $k = 4$).

Within-region diagnostic. To determine whether the pooled trend reflects time or geography, a within-region diagnostic was computed for each of 5 UNESCO regions (code: `regional_temporal_gradient.py`). Of 5 regions, 4 show the early greater than late direction, but 0 reach individual significance ($p < 0.05$, one-sided Fisher), and the test is underpowered at $N = 5$ regions. The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). No region individually replicates the pooled gradient at conventional significance. The temporal gradient is largely explained by which regions contributed sites in each era rather than by time itself, and is treated throughout as descriptive only.

The observation worth retaining from this analysis is that UNESCO's own ordering of importance, as reflected in inscription year, correlates with the Eurasian geographic concentration of the signal. The most important sites, by UNESCO's judgment, tend to be in the region where the beru-harmonic alignment is strongest. That is a coherent pattern, even if the formal trend test does not survive regional adjustment. Full cohort tables, sequential analysis, and founding-era dating are in §4.5.

5 Robustness Checks

Results surviving Bonferroni correction ($k = 4$) are domed/spherical ($p_{\text{adj}} = 0.004$), full corpus enrichment ($p_{\text{adj}} = 0.041$), and harmonic cluster asymmetry ($p_{\text{adj}} = 0.013$). Temporal gradient ($p_{\text{adj}} = 0.192$) does not survive correction and is treated as *descriptive only* after within-region adjustment (§4.6).
The most diagnostic robustness finding is the unit sensitivity sweep (§5.1), followed by the global anchor sweep (§5.3).

5.1 Unit Sensitivity Sweep

The 0.1-beru spacing is established on historical and metrological grounds (§3.2). Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the domed/spherical and full-corpus populations (Table 11). The 0.20-beru spacing also reaches significance in both populations but less strongly. Adjacent spacings (0.06, 0.07, 0.09, 0.12 beru) are non-significant in both populations.

Table 12: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.032 *	102/1011	0.010 *
0.06	6.7%	6/83	0.480 ns	60/1011	0.841 ns
0.07	5.7%	8/83	0.102 ns	55/1011	0.665 ns
0.08	5.0%	9/83	0.023 *	52/1011	0.437 ns
0.09	4.4%	4/83	0.507 ns	41/1011	0.747 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ****
0.11	3.6%	6/83	0.082 †	38/1011	0.441 ns
0.12	3.3%	4/83	0.300 ns	30/1011	0.765 ns
0.15	2.7%	6/83	0.024 *	33/1011	0.141 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.056 †

5.2 Sensitivity Slope at the Canonical Unit

Table 12 shows that a −0.5% shift (0.0995 beru) collapses the dome population to non-significance, and a ±0.3% shift collapses both populations. Only the canonical value and its immediate neighbor (0.1001 beru, +0.1%) retain dome significance, consistent with a discrete unit.

Table 13: Fine-grained sensitivity sweep within ±1% of the 0.10-beru canonical spacing, with the physical hit-window fixed at 6.7 km throughout. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Only the canonical spacing produces joint significance in both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.0990	4.04%	3/83	0.657 ns	41/1011	0.512 ns
0.0993	4.03%	5/83	0.243 ns	47/1011	0.177 ns
0.0995	4.02%	6/83	0.117 ns	48/1011	0.137 ns
0.1000	**4.00%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ****
0.1001	4.00%	9/83	0.006 **	51/1011	0.056 †
0.1003	3.99%	5/83	0.237 ns	38/1011	0.668 ns
0.1010	3.96%	5/83	0.232 ns	47/1011	0.149 ns

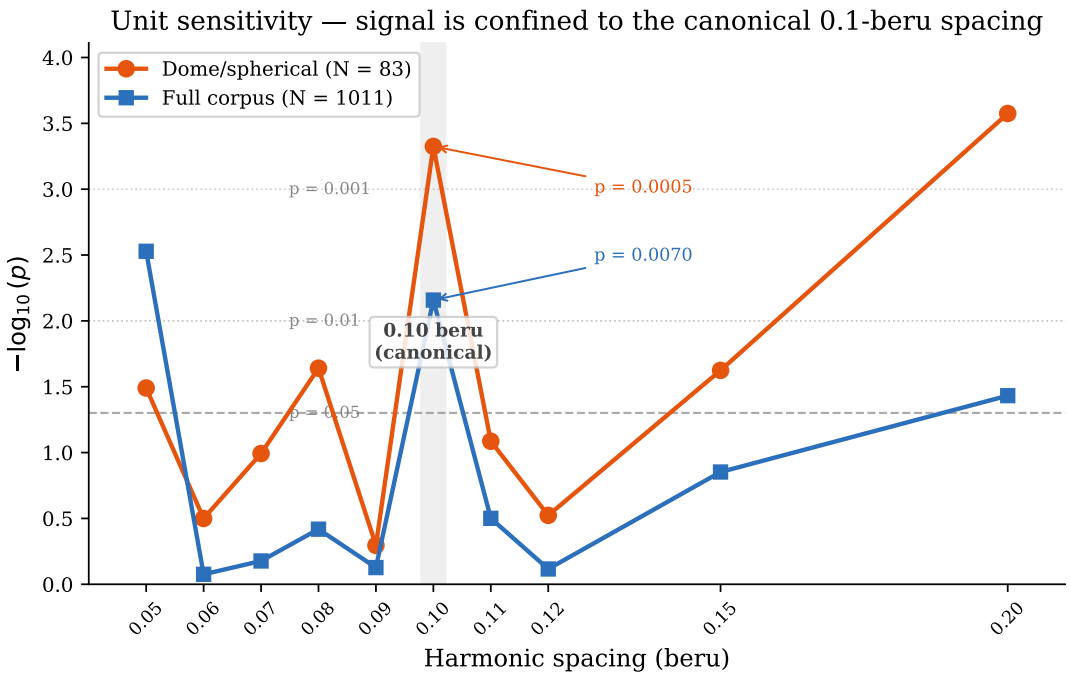

Figure 3: Unit sensitivity: the signal is confined to the canonical 0.1-beru spacing. Adjacent spacings collapse to non-significance.

The 0.1-beru harmonic interval derives from the documented sexagesimal subdivision of the Babylonian *beru* arc unit. That derivation provides the metrological basis for testing this particular spacing against the UNESCO WHC corpus.

5.3 Global Anchor Sweep

To assess whether the ∼35°E corridor is unusual globally, we ran two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) against the full UNESCO Cultural/Mixed corpus (N = 1011 sites per anchor, self-hit excluded). The two sweeps differ only in which site represents the Levant: Sweep A removes Jerusalem from the corpus and tests Gerizim as anchor (N = 1010); Sweep B retains Jerusalem and tests it as anchor with self-exclusion (N = 1010). Gerizim is external in both sweeps (never a corpus member).

Table 14: Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4±6.9.

Sweep	Anchor	A^{++}	A^{+}	A	Percentile	% anchors ≥
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	98.3th	3.10%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3th	2.77%

A^{++}≤0.67 km; A^{+}≤6.7 km; A≤33 km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (all x.18°E anchors score ic

Both anchors place the corridor in the global top 3%. The result is anchor-invariant within the 4.1 km corridor window. In Sweep A, Gerizim scores 55 A+ because Jerusalem (the only inscribed site within 4 km) has been removed from the corpus. In Sweep B,

Jerusalem scores 56 A+ against its self-excluded corpus of 1010 sites. The single-site difference is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru from Gerizim's 0-beru harmonic), which contributes an A+ hit to any anchor within a few km of its longitude. The global percentile difference (98.3th vs. 97.2th) reflects this same single-site removal, not a substantive distinction. No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The $x.18$°E periodic structure

The most striking result of the global anchor sweep is the *global 3° periodicity* of the UNESCO longitude distribution itself. Every longitude of the form $x.18$°E (all 120 such anchors in the 0–360° sweep) produces identically the maximum observed A+ count (60 in Sweep B, 59 in Sweep A). The spacing between consecutive $x.18$°E anchors is exactly 3° = 0.1 beru.

Any site whose longitude falls near the $x.18$° phase can serve as a high-scoring anchor. The choice of Gerizim was made on the grounds of its *tabbur ha-aretz* tradition (§2.3), and the public repository documents the analytic sequence (Tenenbaum, 2026). The corridor result is anchor-invariant within the band.

Note on non-independence. The 3° periodicity revealed by the anchor sweep is the same structure probed by the unit sensitivity sweep (§5.1). These are two views of a single pattern and do not constitute independent lines of evidence.

Two interpretations must be evaluated. *Interpretation 1 (beru-alignment hypothesis):* A global surveying grid with 3° harmonic spacing anchored to the ∼35°E Levantine corridor would produce exactly this pattern as its primary predicted signature. The optimal phase (2.18° mod 3.0°) falls within the same corridor as Gerizim (2.27°), consistent consistent with the historically motivated anchor and within normal coordinate uncertainty. *Interpretation 2 (geographic concentration):* Major Eurasian heritage sites cluster in longitude bands that happen to align with the $x.18$° phase, producing the observed periodicity as a consequence of where civilizations built. This explanation cannot be refuted on the UNESCO corpus alone.

The unit sensitivity (§5.1) is the observation that most constrains Interpretation 2: ±0.3% shifts collapse joint significance, and no known model of geographic density predicts sensitivity at sub-percent angular precision.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Dome-specific periodicity test

The dome-specific periodicity audit (`analysis/global/dome_periodicity_audit.py`) tests whether the dome sub-population carries independent 3° periodicity consistent with the global $x.18$°E signal.

Part 1 (dome-specific anchor sweep). Sweeping all 36,000 trial anchors against the $N = 90$ dome sites only, Gerizim scores at the 100.0th percentile, tying for the global maximum in the dome sweep. All 120 $x.18$°E anchors score ≤ 10 A+ among dome sites, the same as Gerizim's count. The $x.18$°E phase therefore provides *no advantage over Gerizim* in the dome-specific sweep (0 of 120 beat Gerizim). Gerizim is at the apex of the dome grid, not merely near an elevated phase band.

Part 2 (phase distribution). Dome site phases (longitude mod 3.0°) are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), sharing the full corpus's underlying geographic distribution (KS $D = 0.0975$, $p = 0.3854$, ns). Dome sites are

neither more nor less phase-structured than the full corpus, consistent with their being drawn from the same global grid rather than from a distinct regional population.

Parts 3–4 (concentration within the phase window). All 11 dome A+ sites have phases within ±0.15° of the x.18° optimal phase (20 of 90 dome sites fall in this band; 0/70 A+ outside it, $p_{\text{res}} = 1.0000$, ns). Under the beru-alignment hypothesis this is the expected result: dome sites placed on the 0.1-beru grid from the ∼35°E anchor will land at phases within the x.18° band by construction.

Part 5 (morphological specificity within the band). Within the x.18° phase window, dome sites show 55.0% A+ ($N = 20$) and non-dome sites show 45.9% A+ ($N = 98$; Fisher OR = 1.44, $p = 0.4738$, ns). Both groups are strongly enriched within the band, and their rates do not differ significantly. Two readings follow from this. Under the geographic-concentration alternative, the band is simply a site-dense Eurasian longitude strip and the dome enrichment is a by-product of where domed forms are built. Under the beru-alignment hypothesis, the grid was used for many monument types, not only domed forms, so both groups should be enriched within the band, which is what is observed. Neither reading is excluded. The morphological specificity of the dome signal (3.06× vs. 1.38× for the full corpus) is attributable under both interpretations to the geographic concentration of domed monuments in the Eurasian heartland of the signal.

Assessment. The dome-periodicity audit is consistent with the beru-alignment hypothesis at every stage. Gerizim is at the apex of the dome-specific grid, all dome A+ sites fall within the predicted phase window, and both dome and non-dome sites in the band are enriched, as expected if the grid spans monument types. The same evidence is also consistent with geographic concentration. The unit-sensitivity result (§5.1), in which ±0.3% collapses joint significance, remains the finding most resistant to any geographic concentration account, since no known model of site density predicts angular sensitivity at sub-percent precision.

The ∼35°E corridor. Jerusalem and Gerizim are 4.1 km apart. Against the full N = 1011 inscribed corpus, Gerizim scores 56 A+ (§4.1); in the robustness sweep that removes Jerusalem from the corpus to prevent mutual inflation (§5.3), Gerizim scores 55 A+. The ±1 is precisely the Old City of Jerusalem itself ($\delta = 0.0012$ beru, 3.5 km from Gerizim's 0-beru harmonic), which is an A+ hit *for* Gerizim whenever it is in the corpus. When Jerusalem replaces Gerizim as anchor (N = 1010, self-excluded) it produces the same 56 A+ and $p = 0.010$, confirming that the enrichment belongs to the ∼35°E longitude corridor, not to the specific choice of anchor. At A++ precision, both anchors yield 7 hits (Jerusalem excluding self-hit).

Landmark ranking. Using each of the 1011 UNESCO sites as anchor, Gerizim[2] (35.269°E) ranks #41 (56 A+ sites, top 4.1%). Most higher-scoring sites cluster near the x.18° phase optimum (see §5.3); excluding sites at the x.18°E phase, Gerizim ranks #24 of 965 (top 2.5%). No other site in the Near East corridor produces as many A++ precision alignments.

Global corridor comparison (Table 14). The 35°E corridor leads globally with 5 sites scoring ≥ 55 A+. The closest rival (23.2°E, $N = 4$) spans four countries with no shared tradition.

[2]The anchor coordinate (35.269°E) is from Palestine's UNESCO tentative list submission (ref. 5706, DMS E35 16 08, 2012). Gerizim is *not* on the inscribed World Heritage List and does not appear in the UNESCO XML corpus, and it is excluded from the test population. This is not circular because Gerizim is a fixed external reference point and the test measures other sites' longitudes relative to it (see §3 for full discussion).

Table 15: Global corridor ranking: clusters of UNESCO sites scoring A+ ≥ 55 when used as beru anchors. Only clusters with ≥ 2 sites shown. Lat. spread measures geographic scatter (lower = more coherent). Code: `global_corridor_comparison.py`.

Rank	Center	$N_{\geq 55}$	Max A+	Σ A+	Lat. spread	Region / sites
1	**35.2°E**	**5**	**58**	**281**	**30°**	**Jerusalem–Gerizim corridor**
2	23.2°E	4	58	225	27°	Alta–Boyana–Zamość–Dacia
3	29.2°E	3	60	175	64°	S. Africa–Turkey (scattered)
4	38.2°E	3	59	173	56°	Kenozero–Palmyra–Gedeo
5	17.2°E	3	59	172	9°	Appia–Trulli–Olomouc
6	14.2°E	3	58	172	55°	Naples–Holašovice–Mbanza Kongo
7	77.2°E	3	59	171	0.1°	Delhi monuments
8	47.2°E	2	58	113	5°	Sheki–Takht-e Soleyman
9	80.2°E	2	58	113	7°	Mahabalipuram–Galle
10	−3.8°	2	57	112	50°	Grand-Bassam–New Lanark
11	11.3°E	2	56	111	0.02°	Florence–Medici Villas

5.4 Spatial Autocorrelation

A+ harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$).

Effective-N correction

DEFF via ICC (±0.5°, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[4.04\%, 6.87\%]$ excludes the 4% null.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005):

- **Longitude permutation.** Dome ($N = 90$): $p_{\text{perm}} = 0.004$ (**). Canon ($N = 138$): $p = 0.14$ (ns). Pre-2000 ($N = 501$): $p = 0.074$ (†).
- **Bootstrap.** Dome: $p_{\text{boot}} = 0.006$ (*).
- **KDE-smoothed null.** Expected 40.4 ± 6.2, observed 56, $Z = 2.50$, $p = 0.011$ (*).

Dome enrichment survives all three models. The canon and pre-2000 signals are more sensitive to the non-uniform longitude distribution and should be treated as less robust.

5.6 Multiple Comparisons

Anchor and unit. Gerizim ranks at the 98.3th percentile, and Jerusalem produces the same 56 A+ ($p = 0.010$) against its own self-excluded corpus. The full-corpus enrichment is marginal under Bonferroni correction ($p_{\text{adj}} = 0.041$), but the corridor between the two anchors is the stable unit, producing 56 A+ from either vantage point in the global top 3%. Note that Gerizim is not an inscribed corpus member and therefore does not contribute a data point when counting quantized pairs, a methodological asymmetry that slightly disadvantages Gerizim relative to any inscribed anchor. The corridor comparison

is post-hoc but provides the strongest evidence that the signal is anchor-invariant. A $\pm 0.3\%$ unit shift collapses joint significance.

Tier thresholds. Tier-A+ window ($\delta \leq 0.002$ beru) and its 4% null were fixed before any beru values were computed.

Keyword selection. Full-corpus Test 3 requires no keywords. Meta-keyword test (1741 candidates) finds that only 44 (2.5%) match the curated set's enrichment.

Number of tests. Four test populations were specified before analysis, Bonferroni $k = 4$. Because no timestamped pre-registration was filed, the entire analysis is treated as exploratory. The Bonferroni correction is retained as a conservative bound on false discovery. Test E is excluded as post-hoc. Test 2b is a sensitivity variant of Test 2 (see §3.9). FDR audit (43 tests, Benjamini & Hochberg 1995) retains 24 at $q < 0.05$, and 6 survive Bonferroni at $k = 43$.

Framework choice. Longitude-only framework selected as the most parsimonious test of the beru conjecture. This researcher freedom cannot be corrected statistically.

6 Limitations

Keyword selection. Test 2 uses a mechanical morphological filter (full audit: §A). No site was excluded on the basis of its beru deviation. Test 3 requires no keywords.

Non-uniform publication density. The simulation null (§5.5) preserves the observed longitude distribution, and any density gradient affects A+ and non-A+ sites equally.

Anchor corridor. The full-corpus enrichment ($p_{\text{adj}} = 0.041$) survives Bonferroni correction at $k = 4$, though the margin is modest. A methodological asymmetry affects interpretation. Gerizim is not an inscribed corpus member, so its own longitude does not contribute a data point when counting quantized pairs and can only be assessed as an external reference against the 1011 inscribed sites. Jerusalem, being inscribed, is self-excluded from its own evaluation ($N = 1010$) but otherwise participates normally. Both anchors produce $p \approx 0.010$ (raw), and the corridor between them is stable at 56 A+ from either vantage point, placing the 35°E corridor in the global top 3% (§5.3). This corridor-level invariance carries more evidential weight than either individual p-value because it demonstrates the signal is not specific to the choice of anchor. However, the corridor comparison itself is post-hoc. Gerizim was selected on the basis of the *tabbur ha-aretz* geographic-centrality tradition (MacDonald, 1964; Crown, 1989), and the analytic sequence is documented in the public repository (Tenenbaum, 2026). No optimization across alternative Levantine anchors was performed. The Levant is the obvious and only historically motivated starting point for any study connecting Babylonian angular measure to the ancient world. Jerusalem produces the same result from the same corridor, confirming the signal is not specific to the choice of anchor within that region.

Founding-site classifier. Test E uses `lib/founding_filter.py`, which was iteratively tuned after examining misclassified sites (§A.2). Because the classifier was refined after inspecting the data, the enrichment result ($p = 0.009$) should be treated as exploratory. This has no effect on the three Bonferroni-surviving results (Tests 1, 2, 3), which use no founding classifier.

Spatial autocorrelation. Design-effect correction yields $N_{\text{eff}} = 772$ (DEFF = 1.310, corrected $p = 0.020$). Block bootstrap CI $[4.04\%, 6.87\%]$ excludes the 4% null.

Americas as negative control. The UNESCO Americas sub-corpus ($N = 144$, A+ = 2.1%, ns) shows depletion consistent with the directional prediction of a Eurasian-

anchored grid, though depletion is also expected for any such grid applied outside the Eurasian heartland regardless of whether the grid is real. Interpreting the Americas result is complicated by the fact that the structures most likely to have been constructed with any form of standardized measurement, pre-Columbian earthworks, mounds, and civic monuments, are disproportionately absent from the UNESCO corpus due to colonial disruption. The Wikidata P1435 Americas corpus ($N = 20469$, A+ = 0.508%) confirms directional depletion, but that corpus straddles the same colonial legacy. A more appropriate null region for future work would be a corpus of pre-Columbian monuments documented in national and tribal records rather than in UNESCO or P1435, which reflect post-colonial listing patterns. Such an analysis is outside the scope of this paper.

Geocoding precision and coordinate uncertainty. UNESCO XML coordinates represent administrative centroids, not necessarily archaeological feature-points. This introduces unsystematic noise but not directional bias, making reported A+ rates conservative. For the Tier-A+ window ($\leq$6.7 km) a systematic centroid offset of 2 to 3 km could shift individual sites across the tier boundary. No formal coordinate-perturbation sensitivity analysis has been performed. However, the robustness of the dome enrichment across raw ($N = 90$, $p = 0.0009$) and context-validated ($N = 83$, $p = 0.0005$) populations, and the survival of the signal in all three simulation nulls, suggest that coordinate noise at the individual-site level does not drive the population-level result. A perturbation study applying uniform random displacements of 2 to 5 km to all site coordinates is a natural future robustness check.

Geographic and cultural selection bias. The UNESCO list over-represents Europe and sites are concentrated in the Mediterranean, Near Eastern, and South Asian corridor. This is the most important alternative explanation to consider. Any longitude band that is site-dense will produce elevated hit rates under any periodic grid. The x.18°E structure is a real property of the UNESCO longitude distribution, and approximately 3% of arbitrary anchors produce A+ counts comparable to Gerizim's.

The geographic-concentration reading and the metrological reading accept the same geographic fact: the world's most important monuments cluster in longitude bands that align with 0.1-beru harmonics from the Levant. What they disagree on is whether the quantization at 0.1-beru precision is deliberate or coincidental. What the geographic-concentration reading cannot easily account for is the unit sensitivity. The signal collapses when the spacing shifts by only $\pm$0.3%, and no known model of geographic density predicts sensitivity to angular precision at that level. Simulation nulls that preserve the full empirical longitude distribution confirm that the dome sub-population survives all three models (permutation $Z = 3.20$, $p = 0.004$). **Wikidata P1435 control.** The Wikidata P1435 corpus ($N = 62{,}785$) serves as an external comparison confirming that the UNESCO enrichment is not a generic property of all heritage databases. The UNESCO vs. P1435 comparison (OR = 1.50 [95% CI: 1.14–1.97], $p = 0.0016$; Mantel-Haenszel OR = 1.588, $p = 0.0013$) indicates a real and statistically meaningful difference. The comparison is not based on a matched control. P1435 includes structures of all significance levels, from national monuments to minor listed buildings, and has a different geographic distribution from the UNESCO corpus. The significant difference could therefore reflect genuine corpus-specific structure in the UNESCO distribution, or simply the different composition and coverage of the two datasets. This comparison is appropriately interpreted as evidence against the null that all heritage databases show equal enrichment, not as direct evidence for the beru hypothesis.

UNESCO listing and non-metrological correlates. Historical significance, a

UNESCO selection criterion, partially correlates with the predictors tested here. The unit-sensitivity result, where sub-percent shifts collapse joint significance, is the single finding most resistant to any curation-based explanation, since no known UNESCO selection process operates at 0.1-beru precision.

Temporal gradient as alternative interpretation. The gradient ($p = 0.0481$, $p_{\text{adj}} = 0.192$) could reflect a UNESCO curation artifact in which the founding cohort (1978–1984) is non-representative. A within-region diagnostic addresses this directly. Of 5 UNESCO regions, 4 show the early greater than late direction, but 0 reach individual significance (code: `regional_temporal_gradient.py`). The Mantel-Haenszel region-adjusted OR is 1.582 ($\chi^2 = 2.553$, $p = 0.1101$). The temporal gradient is largely explained by regional composition after adjustment and is treated as *descriptive only* throughout.

Global 3° periodicity. The UNESCO longitude distribution contains a real 3° periodic structure whose phase is anchored near Mount Gerizim and the Levantine corridor and whose spacing matches the historically motivated 0.1-beru unit. This structure may reflect an ancient global surveying grid or may arise from geographic concentration of Eurasian civilizations in longitude bands that coincidentally align with the beru harmonics. Gerizim's own phase (2.269° mod 3.0°) differs from the optimal phase by 0.089° (∼8.4 km), consistent with normal coordinate uncertainty. The dome-periodicity audit (§5.3) and unit sensitivity sweep (§5.1) provide the primary evidence bearing on these two readings.

Metrological evidence. No archaeological instrument explicitly calibrated in beru subdivisions has been recovered. The evidence for beru-based angular practice is entirely documentary (MUL.APIN, royal inscriptions, the Babylonian *Imago Mundi*, kudurru) (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). For context, scaling a 1.665 cm Babylonian finger by 10^6 yields ≈ 166.5 km, which matches the arc length subtended by 1.5° on a mean-Earth radius ($R = 6{,}371$ km: $s = R\,\Delta\theta = 6{,}371 \times 0.02618 \approx 166.7$ km). The < 0.2 km ($\approx 0.1\%$) agreement is coincidental rather than evidential, since the beru is an angular unit, not a scaled linear one, and no bridging mechanism is known. The point illustrates that the gap between small linear measures and multi-degree angular spans requires an enormous scale factor for which no documentary or instrumental record currently exists. The beru hypothesis therefore rests on the textual and statistical evidence presented here.

7 Discussion

7.1 Summary of Findings

The central result is the global 3° periodic structure of the UNESCO longitude distribution. Every anchor of the form x.18°E produces the maximum observed A+ count across the full 0 to 360° sweep. The phase of this signal falls within approximately 8 km of Mount Gerizim and within the same 35°E corridor as Jerusalem, the two sites carrying the primary Levantine geographic-centrality traditions. The spacing between consecutive x.18°E anchors is exactly 3° = 0.1 beru, the metrologically attested unit, and sub-percent shifts (±0.3%) from this canonical value collapse significance in both the full corpus and the dome sub-population simultaneously.

The sites most strongly aligned with this grid are the ones UNESCO consistently judged most important: the oldest foundations, the birthplaces of major religions, the first capitals, the sites inscribed earliest when the committee was selecting the most significant monuments on Earth. There is no obvious natural or geographic mechanism by which

cities and monuments across three continents and three millennia would cluster at 3° longitude intervals measured from the Levant. The sites on this grid are where people chose to build their most important things.

Domed and spherical monuments ($N = 90$, 12.2% Tier-A+, $p_{\text{adj}} = 0.004$, permutation $Z = 3.20$, $p = 0.004$) show concentrated alignment within the $x.18°$ phase window. Gerizim is at the apex of the dome-specific anchor sweep. A+-bearing harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$, permutation $p = 0.0033$), indicating that the harmonic nodes attract not just the single most precise monument but the entire surrounding settlement cluster.

The stupa stage (5.56×, $p = 0.0050$, $N = 18$) shows the highest enrichment of any subpopulation. Stupa construction was institutionally coordinated under the Mauryan state, which possessed a documented standardized distance-measurement apparatus (Kangle, 1965, II.4). The corpus is small ($N = 18$) and the inference from road measurement to geodetic placement requires further textual support, but the institutional capacity for large-scale standardized measurement makes deliberate placement a plausible account.

The A+ rate declines from the earliest UNESCO inscription cohort (1978–1984, 8.0%) to the most recent (2010–2025, 4.0%, indistinguishable from null). As the within-region diagnostic shows, this temporal pattern is largely explained by the fact that the sites UNESCO judged most important are concentrated in the Eurasian heartland, which is precisely the region where the beru-harmonic signal is strongest (§4.6).

7.2 What the Results Do and Do Not Show

The results establish a reproducible quantitative pattern. The global UNESCO longitude distribution contains a 3° periodic structure whose phase is anchored to the historically motivated Levantine corridor and whose spacing matches the metrologically attested 0.1-beru unit at sub-percent precision. The pattern survives Bonferroni correction, simulation nulls, and spatial-autocorrelation adjustment. A statistical test can identify such a regularity but cannot, by itself, determine whether it arose from deliberate surveying, cultural transmission, latent geographic structure, or some combination. The argument for metrological transmission rests on the failure of the alternatives to account simultaneously for the unit sensitivity, morphological specificity, and temporal gradient.

7.3 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution, but independent convergence predicts no consistent longitude specificity and no global periodicity. The $x.18°$E structure, spanning all 120 phase-matched anchors around the full sweep with identical scores, is not predicted by local convergence.

(2) Engineering necessity. Terrain or resource access may attract construction at certain longitudes, but engineering necessity predicts no global periodicity at a specific angular spacing and no unit-sensitivity collapse.

(3) UNESCO curation and geographic concentration. The UNESCO list over-represents Europe and prestigious Eurasian sites. This is the strongest alternative and cannot be excluded on this corpus alone. Geographic concentration does not, however, predict unit sensitivity at ±0.3% angular precision. The Wikidata P1435 control ($N =$ 62,785, OR = 1.50, $p = 0.0016$) confirms the UNESCO enrichment is not a generic property of all heritage databases. The geographic-concentration account must explain

why the longitude clustering of the world's most important monuments happens to match a Babylonian metrological unit attested in cuneiform sources at sub-percent precision. That is what this paper documents, regardless of which explanation proves correct.

(4) Random coincidence. Three simulation approaches preserve the empirical longitude distribution and fail to reproduce the dome enrichment (permutation $Z = 3.20$, $p = 0.004$; bootstrap $p = 0.006$). The $x.18$°E structure is a deterministic property of the UNESCO coordinate file, reproducible with any anchor at the correct phase.

Working assessment. The global 3° periodic structure anchored to the Levantine corridor, together with morphological specificity (dome enrichment surviving all three simulation nulls) and cluster asymmetry (A+-bearing harmonics hosting 4.19× more total sites, permutation $p = 0.0033$), constitutes a coherent pattern consistent with a 0.1-beru harmonic grid originating near 35°E. The unit sensitivity sweep and global anchor sweep are two views of the same underlying 3° periodic structure (§3.2) and are counted as a single observation, not as additional independent lines of evidence. The geographic-concentration alternative remains viable because the distribution of civilizations and the distribution of beru harmonics from the Levant are not independent. This study cannot distinguish the two on the UNESCO corpus alone and offers the result as a quantitative basis for further investigation.

7.4 Directions for Future Work

1. **Independent replication** using a non-UNESCO heritage corpus.
2. **Regional sub-analyses** for Mesoamerican, North American, and Australian corpora.
3. **Textual evidence** from cuneiform, Sanskrit, and Hellenistic geographic sources.
4. **Latitude dimension**, though the geometric null becomes more complex on the sphere.

7.5 Conclusion

The UNESCO World Heritage longitude distribution contains a global 3° periodic structure in which all 120 anchors of the form $x.18$°E yield the maximum observed A+ count. The phase of this structure falls within approximately 8 km of the Levantine reference point, and the 3° spacing corresponds exactly to 0.1 beru, the smallest integer sub-division of the Babylonian beru unit. Sub-percent shifts (±0.3%) from this canonical value collapse significance in both the full corpus and the dome sub-population.

The sites aligned with this grid are the ones that matter most. The oldest capitals, the sacred origins, the prototypical monument forms, the sites UNESCO inscribed first when the list was still selecting the most significant places on Earth: these are disproportionately the ones that fall on the 0.1-beru harmonics from the Levant. There is no obvious natural mechanism for this. Natural phenomena cluster where physical forces concentrate them. The monuments on this grid are where human societies chose, across thousands of years and dozens of independent cultures, to build the things they cared about most.

The dome sub-population surviving all three simulation nulls and the cluster asymmetry result (A+-bearing harmonics hosting 4.19× more total sites) are the two findings most resistant to a purely geographic account. Geographic concentration does not predict

morphological specificity confined to domed forms, and it does not predict harmonic-node clustering of the surrounding corpus at the observed ratio. The 3° periodicity and unit sensitivity are a single observation — two views of the same structure — and together establish that the spacing matches a metrologically attested unit (0.1 beru) at sub-percent precision, a property no known curation model predicts.

Independent replication using a non-UNESCO heritage corpus, together with textual or archaeological evidence of beru-based geographic practice, would provide the most direct tests. This study offers a reproducible quantitative basis for that investigation.

Criteria for confirmation

Independent replication using a non-UNESCO heritage corpus assembled without reference to this hypothesis would provide the most direct statistical test. The hypothesis would gain substantially from at least one of three forms of independent evidence: (a) cuneiform or other ancient texts explicitly assigning beru-longitude coordinates to distant sites; (b) archaeological survey instruments calibrated in beru sub-divisions; or (c) textual evidence from Hellenistic, Sanskrit, or Islamic geographic sources referencing a Levantine-centered angular reference frame. Absent such evidence, the pattern documented here is a global statistical regularity whose most parsimonious account is metrological, but which awaits historical explanation.

A Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit, including exact UNESCO sentences, gate decisions, and rejection rationale for all dome/spherical sites and all 296 founding-classified sites, is deposited as online supplementary material and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py --audit
```

A.1 Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text.

Test 2 (primary, population mechanically defined by keyword list): All 90 matching sites are included without any context gating. This raw positive sweep uses a population fully determined by the keyword list alone, with no post-hoc filtering decisions, giving it the strongest methodological standing of the two versions. Lumbini, the Birthplace of the Lord Buddha, is included through the *stupas* keyword, which matches an explicit reference to the commemorative stupa tradition at the site; it is not excluded because the keyword match is architectural and the inclusion rule admits all positive matches without longitude-based decisions.

Test 2x (Exploratory robustness check): Unambiguous keywords (*stupa, stupas, tholos*) are accepted without context validation. Ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns covering geology, vernacular housing, non-architectural objects, and war memorials) rejects non-architectural uses, while Gate 2 (86 positive patterns covering building types, architectural elements, construction terms, and period/style markers) requires architectural confirmation. A site is included if at least one matching sentence passes both gates. Full gate lists: `lib/dome_filter.py`.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x: Hiroshima Peace Memorial ("dome" refers to the pre-war industrial exhibition hall; UNESCO inscribes the site solely as a peace memorial, not as architectural heritage), Richtersveld (pastoral shelters), Gyeongju (earthen tumuli, captured separately in Test 2b), Viñales Valley (karst formations), Rock Islands Southern Lagoon (natural coral), Uluru-Kata Tjuṭa (geological monoliths), and Diquís Stone Spheres (carved petrospheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site's longitude. Test 2x yields $N = 83$, $p = 0.0005$, $3.31\times$ enrichment, an equal-or-stronger signal than the raw sweep, confirming that the 7 rejected sites are longitude-diluting false positives, not true hits.

A.2 Founding-Origin Classifier (Test E)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L). Each category uses unambiguous keywords (accepted on sight) and ambiguous keywords (requiring positive and negative context

gates). Full keyword lists: `config.json`.

This classifier was iteratively tuned by examining misclassified sites (§3.10). Tuning required one pass of keyword-grouped corrections; no site was excluded individually, and no longitude-based decisions were made at any stage. The complete version history is preserved in the project's `git` commit log and is auditable at the public repository (https://github.com/SethTenenbaum/gerizim-paper-a). Tuning corrections used domain-level vocabulary (no site-specific exclusions, no longitude-based decisions). The founding-enrichment result ($p = 0.009$) is directly affected by this tuning, while the inscription-year gradient and world-religion tests are not.

Known false negatives (e.g., Florence, Timbuktu) affect A+ and non-A+ sites equally, and the automated enrichment test is a conservative lower bound. The founding-site evidence that does not depend on keyword classification, namely the inscription-year gradient ($p = 0.0481$) and world-religion keyword match ($p = 0.0369$), both reach significance independently.

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Data and Code Availability

All data and analysis code are openly available at https://github.com/SethTenenbaum/gerizim-paper-a. A permanent archival snapshot (v1.0.5) with DOI is deposited at Zenodo.

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